

RED SEAL EXAM PREP

420B Ironworker (Generalist)

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. What is the purpose of "stirrups" or "ties" in a reinforced concrete beam or column?

- A) To position the longitudinal bars at the correct concrete cover depth
- B) To provide anchorage for the formwork
- C) To resist shear and confine the concrete core
- D) To resist longitudinal tension forces along the beam axis

Q2. An ironworker is placing shear studs on a composite steel beam with a drawn-arc stud gun. Studs from the run are bend tested as a check on weld quality. One stud is bent about 30 degrees off its axis and a crack opens in the weld at the base of the stud. What is the required action?

- A) Accept it - minor bend-test cracks are normal
- B) Replace the stud and raise the test rate
- C) Cut out and re-weld only the cracked stud base
- D) Reject the beam and replace every stud on it

Q3. On a structural drawing, what does the note "FSBW" refer to?

- A) Field Shop Bolt Work
- B) Full-Strength Butt Weld
- C) Fillet Strength Below Weld line
- D) Fabricated Steel Bearing Wall

Q4. What is the purpose of a base plate in a steel column installation?

- A) To prevent corrosion of the column base
- B) To hold the anchor bolts
- C) To allow lateral adjustment after erection
- D) To spread the column load over the foundation

Q5. An opening is left in a steel deck for a duct and, instead of a guardrail, the crew covers it. Under Ontario's Construction Projects Regulation, which of these must that covering satisfy?

- A) It is inspected and certified by an engineer before use
- B) It is used only where a guardrail cannot be installed
- C) It is left liftable so materials can be passed through
- D) It is fastened so it cannot be moved off the opening

Q6. Standard rebar ties are made using:

- A) Epoxy adhesive at each bar crossing
- B) Thin plastic zip ties rated for 50 kg
- C) 16-16.5 gauge soft annealed steel wire
- D) Spot welding at each intersection

Q7. When installing high-strength bolts using the turn-of-nut method, what is the first step?

- A) Install all bolts finger-tight only
- B) Apply lubricant to all bolt threads
- C) Bring all bolts to snug-tight first
- D) Apply full final torque immediately

Q8. What is a spreader beam used for in rigging?

- A) To measure the balance point of a long structural member
- B) To spread two crane boom sections for tandem lifts
- C) To counterbalance the weight of the crane boom
- D) To distribute the load over several pick points

Q9. What does the abbreviation "TOS" mean on a structural steel drawing?

- A) Top Of Steel — the elevation of the top surface of the steel beam
- B) Total Overall Span — the beam span between columns
- C) Tension On Steel — a notation for tension member design
- D) Type Of Section — indicating the steel section type

Q10. An ironworker inspects a 6×19 IWRC wire rope sling before a critical lift and finds 11 randomly distributed broken wires in one rope lay. What action should be taken?

- A) Remove the sling from service immediately
- B) The sling may be used if the broken wires are not load-bearing
- C) Tag the sling for monitoring and continued use
- D) The sling may be used — the limit is 20 broken wires per rope lay

Q11. What does a "flag" on a welding symbol indicate?

- A) The weld is to be made in the field (field weld)
- B) All welds on the drawing must be made in this location
- C) The weld must be inspected by UT (ultrasonic testing)
- D) The weld is a complete joint penetration weld

Q12. On an Ontario construction project a worker has taken a fall while wearing a personal fall arrest harness. What is the correct procedure for the harness and lanyard?

- A) Only the lanyard needs to be replaced after a fall; the harness can be reused
- B) If the worker is unhurt, the harness can be put back in service immediately
- C) Perform a 10-minute visual inspection and return to service if no visible damage
- D) Remove from service; reuse only if the manufacturer certifies it safe

Q13. On a structural drawing, the connection dimensions for a diagonal brace are run from a mark labelled W.P., the working point. Unless the detail states otherwise, what does that mark locate on the joint?

- A) The corner of the gusset plate nearest the column face
- B) The point where the brace edge crosses the beam top flange
- C) Where the centrelines of the connected members intersect
- D) The centre of the bolt group joining the brace to the gusset

Q14. During a concrete pour on a steel-framed floor, an ironworker sees the decking buckling and the frame deflecting far more than expected. What is the correct immediate action?

- A) Stop the pour, clear the area below and notify the engineer
- B) Relocate the pump hose to spread the load and carry on pouring
- C) Treat it as normal elastic deflection and keep placing concrete
- D) Speed up the pour so the load sits on the deck for less time

Q15. ASTM F3125 Grade A325 and Grade A490 are the two common high-strength structural bolt grades. Which statement about them is correct?

- A) A490 may be reused after full pretensioning; A325 may not
- B) Both require the same minimum pretension at a given size
- C) A325 is stronger than A490 and used for most connections
- D) A490 requires a higher minimum pretension than A325 does

Q16. A complete joint penetration (CJP) groove weld requires:

- A) Weld metal to fill the groove at least halfway
- B) Post-weld heat treatment regardless of steel grade
- C) Weld metal penetrating the joint's full thickness
- D) At least three weld passes in the groove

Q17. A rigid steel weldment is picked on a four-leg bridle sling. Under British Columbia's OHS Regulation, the working load limit of the whole sling is limited to the working load limit of how many of its legs?

- A) Two of them, the pair on the same diagonal
- B) Whatever number a professional engineer sets
- C) All four, since the four legs share the load
- D) Three of them, whichever three are chosen

Q18. What is the effective throat of a 12 mm fillet weld?

- A) 6 mm
- B) 8.5 mm
- C) 10 mm
- D) 12 mm

Q19. What is an "ironworker machine" (also called a hydraulic ironworker)?

- A) A hydraulic torque wrench for high-strength bolt installation
- B) A mobile crane specifically designed for steel erection
- C) A portable wire rope swaging machine for rigging
- D) A hydraulic machine that shears, punches, and bends steel

Q20. What does "100% tie-off" mean for ironworkers during structural steel erection?

- A) The ironworker must be tied off for 100% of the workday
- B) 100% of the rigging must be secured before erection begins
- C) The ironworker must use a full 100-foot lifeline at all times
- D) At least one PFAS connection at all times, even when moving

Q21. What does a "Personal Fall Arrest System" (PFAS) consist of?

- A) Safety net and horizontal lifeline only
- B) Safety boots, hard hat, and safety glasses
- C) Body belt, rope grab, and vertical lifeline
- D) Full body harness, connecting lanyard, and anchor point

Q22. What does "plumbing up" a steel structure mean?

- A) Applying anti-corrosion primer to exposed steel
- B) Installing the water supply lines before steel is covered
- C) Installing floor deck drainage systems
- D) Adjusting columns and beams to vertical and level

Q23. In structural steel work, what does CSA W59 govern?

- A) The certification of individual welders for structural work
- B) The testing and classification of welding consumables
- C) The safety requirements for welding fumes and gases
- D) The design and fabrication of welded steel structures

Q24. A hydraulic bolt tensioner is used instead of a torque wrench for high-strength bolt installation. What is the key advantage?

- A) It works on all bolt grades including stainless steel and aluminum
- B) It applies direct tension, eliminating torque variability
- C) It is lighter and cheaper than a torque wrench
- D) It can install bolts in confined spaces where a torque wrench cannot fit

Q25. Under AWS D1.1, when welding ASTM A572 Grade 50 steel with low-hydrogen SMAW electrodes, the minimum prequalified preheat and interpass temperature for material over 38 mm up to 65 mm thick is approximately:

- A) 107°C (225°F)
- B) No preheat required
- C) 66°C (150°F)
- D) 10°C (50°F)

Q26. Which crane hand signal means "STOP"?

- A) Arm extended, palm facing down, moving arm side to side
- B) Both arms raised above head
- C) One arm raised, rotating at the elbow
- D) Arm extended, palm facing the crane operator, fist closed

Q27. Rebar chairs (bar supports) are used to:

- A) Hold rebar at the correct height for cover
- B) Hold vertical rebar in position while tying
- C) Mark the location of rebar on the concrete surface after pouring
- D) Splice rebar together at joints

Q28. Rebar arrives on site carrying tight mill scale and light surface rust. What should the ironworker do before it is placed?

- A) Apply a rust inhibitor coating before placing in forms
- B) Place the bars as is — light rust is acceptable
- C) Replace all lightly rusted bars with new material
- D) The bars must be sandblasted clean before placement

Q29. A crane is lifting a 20,000 kg load at a 15m radius. The load chart shows maximum lift capacity at 15m is 18,000 kg (with all outriggers fully extended on firm ground). What should the ironworker signalperson communicate to the operator?

- A) Proceed if the crane manufacturer approves in writing
- B) Proceed if the lift will be completed in under 2 minutes
- C) Proceed — crane charts include a 10% safety margin for occasional overloads
- D) Stop the lift — it exceeds the chart capacity

Q30. Two structural steel beams are to be connected to a common column. Under Ontario's Construction Projects regulation, how may that connection be made?

- A) With both beams sharing one set of bolts through the column
- B) As a clipped double connection, or the first beam seated
- C) With the second beam tack welded until the bolting crew comes
- D) With the first beam's bolts pulled to let the second one in

Q31. What is the purpose of shear studs welded to the top flange of a steel beam?

- A) To increase the buckling resistance of the top flange
- B) To provide connection points for safety lanyards
- C) To attach the metal deck to the beam
- D) To create composite action with the concrete slab

Q32. A weld inspection reveals "undercut" along the toe of a fillet weld. What caused this?

- A) Excessive amperage or travel speed
- B) Insufficient preheat temperature
- C) Electrode too small for the fillet size
- D) Moisture in the electrode coating

Q33. Under the Alberta OHS Code, what is the maximum free-fall distance permitted for a personal fall arrest system that has no shock absorber?

- A) 0.6 m (2 ft)
- B) 3 m (10 ft)
- C) 1.2 m (4 ft)
- D) 1.8 m (6 ft)

Q34. During a crane lift, what is the purpose of a tagline?

- A) To control the rotation and swing of the load
- B) To measure the length of the lift path
- C) To attach the load to the crane hook
- D) To provide an extra rigging connection in case the main sling fails

Q35. A wire rope sling is rated for a working load limit (WLL) of 10,000 kg in a straight-line (vertical) hitch, and the sling tag shows no separate choker rating. What is the effective WLL in a choker hitch at an angle of choke of 120° or greater?

- A) 5,000 kg — choker hitch reduces capacity by 50%
- B) 10,000 kg — choker hitch does not reduce capacity
- C) 8,000 kg — choker hitch reduces capacity by 20%
- D) 7,500 kg — choker hitch reduces capacity by 25%

Q36. A column is being lowered onto its anchor rods. What is the correct practice for the connectors receiving it?

- A) Steady the base plate by hand so the holes drop over the rods
- B) Guide it with taglines, hands and feet clear of the base plate
- C) Foot the base plate from the footing to stop it from swinging
- D) Slack the load line so the column settles onto the rods freely

Q37. On a structural drawing, what does the symbol "EL. 12500" typically mean?

- A) The point's elevation above the project datum
- B) Element number 12,500 on the fabrication list
- C) The element has a design load of 12,500 kg
- D) The element is 12,500 mm in overall length

Q38. An anchor rod at a column base is 25 mm out of position. Which statement about correcting it in the field is correct?

- A) Reaming the base plate hole oversize keeps the rod's capacity
- B) Cutting it off and welding on an extension is a standard field fix
- C) Bending it cold with a rod bender restores its design strength
- D) Heating or flame-cutting the rod changes its strength properties

Q39. An explosive actuated fastening tool misfires while an ironworker is shooting deck fasteners, and the load does not go off. Under Ontario's Construction Projects regulation, what happens to that load?

- A) It goes back into the load container with the unused loads
- B) It goes into the day's construction waste at the end of shift
- C) It goes into a water-filled container on site until removed
- D) It is re-chambered and fired into scrap steel to spend it

Q40. No manufacturer's specification can be found for a winding drum that a wire rope will spool onto. What minimum drum diameter should be used?

- A) 20 times the rope diameter
- B) 6 times the rope diameter
- C) 30 times the rope diameter
- D) 12 times the rope diameter

Q41. What is a "lap splice" in reinforcing steel?

- A) A welded joint fusing two rebar pieces at their ends
- B) A mechanical coupler that joins two rebar end-to-end
- C) An overlap where two parallel bars transfer stress
- D) A bar bent into a U-shape hooked around another bar

Q42. A structural drawing shows "4-M20 A325 bolts" at a beam-to-column shear tab connection. What does this mean?

- A) A bolt group with a shear capacity of 4,000 N per M20 bolt
- B) Metric bolts, Group 4 size class, rated 20 kN shear capacity each
- C) 4 machine bolts of 20 mm diameter, Grade A325 galvanized finish
- D) 4 bolts, metric M20 (20 mm diameter), ASTM A325 high-strength

Q43. Carbon arc gouging is used in structural ironwork primarily for:

- A) Cutting plate to exact dimensions in the fabrication shop only
- B) Welding thick plates that require deep penetration
- C) Preheating steel before welding
- D) Removing defective welds and back-gouging root passes

Q44. An ironworker on a steel beam is tied off to a self-retracting lifeline anchored overhead. Before work starts, the fall clearance needed below the working surface must be worked out. Which distances are added together?

- A) Only the arresting force rating divided by the total worker weight
- B) Free fall before lockup, deceleration, D-ring to feet, safety margin
- C) Only the length of lifeline paid out when the worker steps off the beam
- D) Nothing, because a retracting lifeline removes the need for clearance

Q45. What is the primary cause of porosity in SMAW structural welds?

- A) Using E7018 electrodes at too low an amperage
- B) Moisture or oil on the metal or electrode
- C) Failure to remove slag between weld passes
- D) Excessive root opening in a groove joint

Q46. A beam detailed with 25 mm of camber arrives on site with its camber paint mark worn off. How does the crew establish which way up it goes before it is flown?

- A) Sight along the beam on the ground to find its natural bow
- B) Measure the depth at both ends and put the deeper end up
- C) Fly it either way, since 25 mm is inside erection tolerance
- D) Assume the mill mark stamped on the web is meant to face up

Q47. Open-web steel joists have been landed and both bearing ends attached, but the bridging is only partly installed. What may be placed on them?

- A) Deck bundles, provided each one is landed over a bearing end
- B) Loads up to half of the joists' design capacity may be placed
- C) Anything the joists were designed to carry, once both ends bear
- D) No construction load at all until the bridging is fully anchored

Q48. An ironworker is rigging a steel plate that has sharp flame-cut edges, and a polyester web sling will bear directly on one of those edges. What has to be done before the plate is lifted?

- A) Double the sling so the added ply takes the cutting action
- B) Fit corner protectors between the plate edges and the sling
- C) Take up the load slowly so the edge cannot saw the webbing
- D) Cut the rated load in half and lift the plate as it is rigged

Q49. What is the difference between a "drift pin" and a "spud wrench"?

- A) They are the same tool — different regional names
- B) A drift pin can tighten bolts; a spud wrench cannot
- C) A spud wrench is used only for rigging; a drift pin is used only for concrete work
- D) A spud wrench adds a wrench end; a drift pin only aligns

Q50. What do the deformations (ribs) on deformed rebar do?

- A) Reduce the bar cross-section to save material at low-stress locations
- B) Mechanically interlock with concrete to improve bond and force transfer
- C) Increase the bar weight to improve resistance to impact loads
- D) Provide identification markings for the steel mill

Answer Key & Explanations

Q1. Answer: C

Stirrups (beams) and ties (columns) resist shear and torsion forces, confine the concrete core to increase ductility, and brace the longitudinal bars against buckling under compression load.

Q2. Answer: B

A crack in the weld during a bend test is a failed test, not a normal result. Bending the stud off its axis loads one side of the weld collar in tension, and a sound drawn-arc weld takes that without cracking. A crack means the weld metal or the fusion at the base is unsound - commonly from moisture, paint, mill scale or oil on the flange, a damaged or damp ferrule, or wrong current or plunge time. The response is to replace the failed stud, removing it and making good the base metal before welding a replacement with corrected settings, and to widen the sample so the studs put down alongside it under the same conditions are proved as well. Dressing and re-welding the base of a suspect stud proves nothing about that stud, and one failure does not condemn every stud on the beam. How many studs are tested, and how far the sampling is widened after a failure, come from the project specification and the welding inspection procedure for the job.

Q3. Answer: B

FSBW (Full Strength Butt Weld) or CJP (Complete Joint Penetration) groove weld indicates that the weld must develop the full tensile strength of the connected base metal — the weld is not the weak link. This requires 100% penetration through the joint thickness, usually verified by visual inspection plus UT (ultrasonic testing) or radiography. It is different from: a partial joint penetration (PJP) weld, which does not penetrate the full thickness and has reduced capacity; a fillet weld, which joins members at a surface (no groove); and a plug or slot weld. FSBW/CJP welds are required at moment connections (beam-column moment joints) and splice plates carrying tension — they cost more (more prep, more passes, more inspection) than fillet welds.

Q4. Answer: D

A base plate distributes the concentrated column load over a larger bearing area on the concrete foundation, reducing bearing stress to within allowable limits.

Q5. Answer: D

O. Reg. 213/91 s. 26.3 (2) gives two ways to stop a worker falling through an opening on a work surface - a guardrail system, or a protective covering - and it sets out what the covering has to be: it completely covers the opening, is securely fastened, is adequately identified as covering an opening, is made from material adequate to support all loads to which it may be subjected, and is capable of supporting a live load of at least 2.4 kilonewtons per square metre. Fastening is the one that site practice most often skips. A sheet lying loose over a hole is moved by the very traffic it is there to protect: kicked aside, slid off as material is dragged over it, or lifted by wind. The worker who then steps on the edge of a shifted cover goes through it.

That is also why a cover left deliberately liftable for access is not a compliant cover. Where work has to be done in or around the opening, s. 26.3 (3) allows the covering or the guardrail to be removed temporarily, but only while the worker is adequately protected and danger signs are posted in accordance with s. 44 (1) and (2) - a planned removal under protection, not a cover that anybody can pick up.

The other two answers borrow rules that belong elsewhere. Inspection and testing by an engineer, or by a competent person under the engineer's supervision, before the system goes into service is the safety net requirement in s. 26.8 (3); nothing in s. 26.3 puts an engineer on a deck cover. And the covering is not a fallback that first needs a guardrail to be impracticable: s. 26.3 (2) sets the two out as alternative precautions for an opening, so a covering that meets all five requirements is protection in its own right. Note also that this duty does not wait on a height - section 26 lists falling through an opening on a work surface as a hazard in itself, alongside falling more than 3 metres.

Q6. Answer: C

Rebar ties are made from 16 to 16.5 gauge (approximately 1.6 mm diameter) smooth, soft annealed steel wire, twisted with a tier tool. Multiple tie patterns exist for different conditions (snap tie, saddle tie, figure-8 tie).

Q7. Answer: C

Turn-of-nut requires all bolts in the connection to be brought to the snug-tight condition first (full effort of an ironworker with a spud wrench), then a specified rotation is applied to achieve the minimum pretension.

Q8. Answer: D

A spreader beam is a below-the-hook lifting device that widens the pick points on a load. The beam carries the spread as compression along its own length, so the sling legs hang nearer to vertical and no horizontal squeeze is passed into the load itself. It does not counterbalance anything on the crane, it does not find a balance point, and it has nothing to do with boom sections. Do not confuse it with an equalizer beam, which is a different device: an equalizer beam pivots at a centre point to share load equally between two hoist lines or two cranes. Alberta's OHS Code s.297(2) puts below-the-hook lifting devices other than slings under ASME B30.20, and s.297(3) and (4) allow a spreader bar to be labelled with its rated capacity by a uniquely numbered capacity data sheet. BC's OHS Regulation requires the working load limit of a spreader bar to be certified by a professional engineer or established by the manufacturer (s.15.58), requires a nameplate (s.15.59), and counts the device as part of the lifted load (s.15.60).

Q9. Answer: A

TOS (Top of Steel) is an elevation reference on structural drawings indicating the height of the top flange of a beam, measured from the project datum. It is used to coordinate with the concrete deck, mechanical, and architectural drawings. Common related notations: BOS (Bottom of Steel — soffit elevation, important for clearance), TOS is NOT the same as finished floor elevation (which is TOS + deck slab thickness + topping if any). Wrong answers often confuse TOS with the beam depth or with the centreline elevation. For steel erection: when setting beam elevation, you level to the TOS — not to the bottom flange or web centreline.

Q10. Answer: A

ASME B30.9, para. 9-2.9.5(b)(1): a strand-laid or single-part wire rope sling must be removed from service at 10 randomly distributed broken wires in one rope lay, or 5 broken wires in one strand in one rope lay. Eleven randomly distributed breaks exceeds that limit, so the sling comes out of service immediately — tag it, segregate it so it cannot be picked up again, and report to the rigger-in-charge. Note what the standard does not say: B30.9 sets no broken-wire limit keyed to rope construction, so there is no separate 6×19 or 6×36 figure. Classification governs the minimum clear length of rope between splices (para. 9-2.3.2), not the wire count. The one construction that changes the count is the cable-laid sling, at 20 broken wires per lay. B30.9 is not merely an American reference here — it is adopted by regulation: Alberta OHS Code Part 21, s.297(1) requires wire rope slings manufactured on or after July 1, 2009 to meet ASME B30.9, and s.306 adds further removal triggers, including wear or corrosion affecting individual wires over more than one-third of the original rope diameter, bird-caging, kinking, and heat or arc damage. Beware of carrying figures across: provincial limits for running rope are different and tighter (Alberta s.306(2): 6 randomly distributed broken wires in one rope lay, or 3 in one strand), but a sling is not a running rope. Check the rigging part of your own province's regulation.

Q11. Answer: A

A flag on the weld symbol reference line indicates a field weld — one that is to be made during erection at the job site, as opposed to a shop weld made during fabrication.

Q12. Answer: D

Ontario O. Reg. 213/91, Section 26.6(8): if a worker using the fall arrest system falls, the system shall be immediately removed from service and shall not be used again by a worker unless all components of the system have been certified by the manufacturer as being safe for re-use. Every component that took the arrest force goes out of service, not just the lanyard - webbing fibres, stitching and hardware can be damaged internally with nothing showing on the outside, which is why a visual check settles nothing. Alberta is built the same way: OHS Code Section 150.2(2) removes a personal fall arrest system from service once it has stopped a fall, and Section 150.2(3) returns it only if a professional engineer or the manufacturer certifies that it is safe to use. A worker's own inspection is the pre-use check required before the equipment is used on each shift; it is not what puts arrested equipment back to work.

Q13. Answer: C

The working point is where the centrelines - the gravity axes - of the members meeting at that joint intersect.

The detailer sets the joint out from that point so that the lines of action of the column, the beam and the diagonal brace all meet in one place. When they do, the joint delivers axial force and no member picks up a bending moment from an offset. Shift the brace's line of action away from the working point and an eccentricity is introduced, and the moment that eccentricity produces has to be carried by the connection and by the members framing into it. The choice of working point belongs to the engineer, so a detail that deliberately works to a point away from the member centrelines says so on the drawing and the connection is designed for that eccentricity - which is why the stem says "unless the detail states otherwise".

For the crew the working point is a layout reference, not a piece of hardware. On a brace-to-beam-to-column joint it commonly falls where the column and beam centrelines cross, inside the members themselves, where there is nothing to fasten to; the gusset plate exists precisely because the members cannot all physically meet at that point. That is what separates it from the three wrong answers, each of which is a real feature you can put a tape on. The centre of the bolt group is where the brace is fastened to the gusset, and its position is set out from the working point rather than the other way round. The corner of the gusset and the place where the edge of the brace crosses the beam flange are edges of fabricated plate and shape; they move if the gusset is resized or the brace section is changed, while the working point stays where the member centrelines cross.

Q14. Answer: A

Unexpected movement during a pour is a warning of a developing failure, and at that moment the cause is not yet known. It could be shoring or falsework under-designed or kicked out, concrete piling up in one place instead of being spread, deck load beyond what the frame was designed to carry, or a connection that has slipped. Working out which one it is while standing on the deck with concrete still going on is guesswork, so the immediate action does not wait on the diagnosis: stop placing concrete, get everyone out from under and beside the affected bay, and call the engineer of record before anything else is done. Moving the pump hose acts on one guess while people remain under a structure that is already behaving abnormally, pouring faster adds load faster, and calling it normal elastic deflection ignores the buckling decking, which elastic deflection does not produce.

Q15. Answer: D

The higher-strength grade carries the higher pretension. Grade A490 has a minimum tensile strength of 150 ksi against 120 ksi for Grade A325, and because the specified minimum pretension is set at 70 percent of the bolt's minimum tensile strength times its tensile stress area, A490 comes out about a quarter higher at every diameter. The RCSC Specification for Structural Joints Using High-Strength Bolts, the United States bolting specification, tabulates it: a 3/4 in bolt is 28 kips for A325 against 35 kips for A490, 7/8 in is 39 against 49, and 1 in is 51 against 64, with A490 higher in every row. The two grades therefore never share a pretension at a given size, and A325 is the lower-strength grade, so it cannot be the stronger one however often it appears on a job. Reuse runs the opposite way from the idea that the higher-strength bolt can be tightened twice. A pretensioned installation stretches the threads between the nut and the thread run-out into the inelastic range, and A490 bolts are not consistently ductile enough to survive a second pretensioning, so RCSC bars any reuse of A490 and of galvanized or coated bolts of either grade; only a plain-finish A325 heavy hex bolt may be reused in a pretensioned joint, and only with the engineer of record's approval. Retightening a bolt that loosened while its neighbours were installed is touching up, not reuse. On Canadian work the design, installation and inspection of bolted joints are governed by CSA S16 for buildings and by CSA S6 or the provincial specification for road bridges; those standards adopt many RCSC provisions but not all of them, and a project specification can be stricter still - Alberta Transportation's Standard Specifications for Bridge Construction has every bolt tensioned once only and permits no reuse at all.

Q16. Answer: C

A CJP weld requires weld metal to completely penetrate the full thickness of the joint, filling the entire cross-section and achieving full fusion. It is required in joints subject to tension, cyclic, or seismic loading.

Q17. Answer: D

WorkSafeBC's OHS Regulation s. 15.33 (2) is one sentence: the working load limit of a sling with more than three legs is limited to the working load limit of any three legs of the sling. A fourth leg is allowed on the hook; it simply cannot be counted. The mechanics behind that are worth carrying to any jurisdiction. Four legs holding one rigid piece are statically indeterminate - there are more supports than the statics of the lift can define - so which legs are actually tight is decided by millimetres of leg length and by exactly where the pick points sit, not by symmetry. Nothing about the arrangement guarantees that all four come up tight together, so adding four leg capacities describes a lift that may not happen. The section does not stop at two, though: three is the cap it states. An engineer's stamp is the answer people reach for when a rule looks severe, but this subsection sets the cap itself and does not hand the number to anyone to reset. Two further limits ride along in the same section: the working load limit of no individual component of the assembly may be exceeded (15.33 (1)), and the load carried by any single leg of a bridle must not exceed the working load limit of that leg (15.33 (3)).

Q18. Answer: B

For a standard fillet weld with equal legs, the theoretical throat = $\text{leg} \times \sin(45^\circ) = 12 \times 0.707 \approx 8.5 \text{ mm}$. The effective throat is used to calculate weld strength.

Q19. Answer: D

An ironworker machine is a multi-function hydraulic bench-top or freestanding machine used in fabrication shops and on site for shearing plate, punching bolt holes, notching angle iron, and bending bar stock.

Q20. Answer: D

100% tie-off means continuous connection — the ironworker must have at least one PFAS connection at all times, including while moving or transitioning their tie-off point. A double-lanyard or SRL with a second connection allows transitioning without a gap.

Q21. Answer: D

A PFAS has three essential components: (1) full-body harness, (2) connecting means (lanyard, self-retracting lifeline, rope grab + lifeline), and (3) an anchor point with sufficient strength.

Q22. Answer: D

Plumbing up involves checking and adjusting the structure so columns are perfectly plumb (vertical) and beams are level before making final bolted or welded connections.

Q23. Answer: D

CSA W59 (Welded Steel Construction — Metal Arc Welding) specifies design, fabrication and erection requirements for welded steel structures in Canada, including joint design, acceptable workmanship and inspection criteria. It is often confused with CSA W47.1. W59 is the construction standard — what must be built and to what quality — while W47.1 certifies a fabricator or erector as a company qualified to do the welding, and CSA W178.2 certifies welding inspectors. Welding consumables belong to the CSA W48 series and welding health and safety to CSA W117.2, so neither of those sits in W59 either. AWS D1.1 is the equivalent American standard; Canadian work is built to W59, and D1.1 governs only where a project specification calls it up. For the Red Seal ironworker exam: W59 is the welded construction standard, W47.1 is company certification, W178.2 is inspector certification.

Q24. Answer: B

Hydraulic tensioners apply direct tensile load to the bolt by pulling the threaded end rather than applying torque, eliminating the torque-to-tension variability that comes from lubrication, thread condition, and technique. This achieves more consistent and accurate pretension.

Q25. Answer: C

AWS D1.1 places ASTM A572 Grade 50 in Category B, and Category B welded with low-hydrogen SMAW electrodes calls for 150°F (66°C) on material over 38 mm through 65 mm thick. The full Category B row of the prequalified preheat table (Table 5.8 in the 2020 and later editions; Table 3.2 in earlier ones) reads: 3 to 20 mm = 32°F (0°C); over 20 through 38 mm = 50°F (10°C); over 38 through 65 mm = 150°F (66°C — the metric column of D1.1M rounds this to 65°C); over 65 mm = 225°F (107°C). So 107°C is correct only for plate thicker than 65 mm, and that is the trap in this question. Watch the band edges too: a section exactly 38 mm thick still falls in the 10°C band, because the row reads 'over 38'. On Canadian structural work AWS D1.1 is not the governing document. Building Code Article 4.3.4.1 requires buildings and structural members made of structural steel to conform to CAN/CSA S16, and S16 in turn requires a CSA W47.1 certified fabricator with joint design and quality to CSA W59 — so preheat comes off the CWB-accepted WPS/WPDS, never off a US table from memory. The Canadian counterpart grade is CSA G40.21 350W (350 MPa minimum yield, against 345 MPa for A572 Grade 50), which is what a Canadian mill certificate will normally show.

Q26. Answer: A

The universal crane signal for STOP is arm extended horizontally with palm facing down, moved side to side. EMERGENCY STOP uses both arms in this position.

Q27. Answer: A

Bar chairs and supports are plastic or metal supports placed under rebar to maintain the required concrete cover by holding the bars at the correct height above the form (distance from bottom of rebar to bottom of slab).

Q28. Answer: B

Tight mill scale and light surface rust are acceptable and need not be removed; light rust may slightly improve bond by roughening the surface. Only loose scale and loose, flaking rust have to come off, and the bar must still meet its specified dimensions, mass and deformation height after hand wire brushing. Mill scale and rust are not the same thing: mill scale is the tight blue-black oxide left by hot rolling, while rust is the orange-brown corrosion product that forms afterwards. Canadian project specifications written to CSA A23.1 call for removal of loose scale and loose rust only.

Q29. Answer: D

Load chart values are maximums — they MUST NOT be exceeded under any circumstances. Crane load charts already include safety factors per CSA Z150. Exceeding the chart rating voids the chart, creates an unacceptable risk of crane overturning, and is a criminal liability. The lift must be stopped and re-planned: decrease the lift radius (boom in), improve ground bearing with outrigger pads, or bring in a larger crane.

Q30. Answer: B

Ontario Regulation 213/91, s. 31 (2) allows two arrangements and no others: if two structural steel columns or structural steel beams are connected to a common column or common beam, the connection shall be made using a clipped double connection, or the first column or beam shall be secured in a seated connection. Both arrangements do the same job - the first member keeps support of its own, so hanging the second member does not depend on disturbing what is holding the first. That is what makes the other answers the wrong instinct. Pulling the first beam's bolts to make room takes the support out from under a member that a connector may be standing on. A tack weld is a fit-up aid rather than the connection the drawings call for, and it is not one of the two arrangements the section permits. Running both beams on one common set of bolts leaves the two members depending on the same fasteners, which is the arrangement the section is written to prevent.

Q31. Answer: D

Shear studs are welded to the beam top flange and embed in the concrete deck above. They transfer horizontal shear forces between the steel beam and slab, creating composite behaviour that significantly increases beam capacity.

Q32. Answer: A

Undercut is a groove melted into the base metal at the weld toe that is not filled by weld metal. It comes from too much heat at the toe — excessive amperage, too fast a travel speed, too long an arc, or an electrode angle that throws the arc onto the vertical member instead of into the corner. It matters because it removes section from the base metal and leaves a sharp notch at the toe, which is where fatigue cracks start. The other faults listed cause different problems: moisture in the coating causes porosity, too little preheat on heavy or restrained sections invites cracking, and too small an electrode gives an undersized weld rather than a gouged toe.

Q33. Answer: C

The Alberta OHS Code requires a personal fall arrest system without a shock absorber to limit a worker's free-fall distance to 1.2 m. A shock absorber is normally required in the first place; a system without one is allowed only where it meets that free-fall limit and the rest of the same section. Where a shock absorber is used, the permitted free fall is the maximum free-fall distance the absorber is labelled and rated for, and the system must still be arranged so the worker cannot strike the ground or a level below, and must limit the maximum arresting force on the worker to 6 kN, or 8 kN with an E6 shock absorber used to the manufacturer's specifications. The practical control on free fall is anchor height: the Code requires the lanyard to be secured to an anchor no lower than the worker's shoulder height, or as high as reasonably practicable.

Q34. Answer: A

Taglines allow workers to guide and control a suspended load from a safe distance, preventing rotation and swing without placing workers directly under or near the load.

Q35. Answer: D

ASME B30.9-2021, para. 9-2.10.1(d): when the choker hitch rating is not identified on the sling, the choker hitch rating for single-leg and bridle wire rope slings is 75% of the sling's straight-line hitch rating (70% for cable-laid slings). So $10,000 \text{ kg} \times 0.75 = 7,500 \text{ kg}$ — a 25% reduction, caused by the bending and pinching of the rope at the choke point. That 75% figure holds only at an angle of choke of 120° or greater: para. 9-2.10.1(e) sends you to Table 9-2.10.1-1 for smaller angles, which gives 87% of the choker rating at 90° – 120° , 74% at 60° – 89° , 62% at 30° – 59° , and 49% at 0° – 29° . Although B30.9 is an American standard, it governs Canadian work because occupational health and safety regulation adopts it by reference — Alberta OHS Code Part 21, s.297(1) requires wire rope, alloy steel chain, synthetic fibre rope, metal mesh and synthetic fibre slings manufactured on or after July 1, 2009 to meet ASME B30.9. If the sling tag carries its own choker rating, use the tag.

Q36. Answer: B

IHSA's Construction Health and Safety Manual chapter on structural steel erection (Ironwork, M036, 2023 edition) gives two controls for column standing: keep the load line vertical and taut while the column is being stood, and keep hands and feet clear of the base plate while the column is being lowered onto the anchor rods. The base plate is the pinch point. A column landing on four rods can shift sideways in an instant, and a hand steadying the plate or a boot bracing it against the footing is between a plate of several hundred kilograms and hard concrete. Control the piece from a distance with taglines instead. Slackening the load line before the column is stable is the other error in this list: the same chapter directs that the column be stable before the rigging is disconnected, and a slack line lets the piece take charge of itself while the crew is still reaching into the connection.

Q37. Answer: A

EL. (Elevation) followed by a number — for example EL. 12500 — gives the finished height of that point in millimetres above the project datum, typically the finished first floor established at EL. 0. So EL. 12500 is 12,500 mm, or 12.5 m, above datum. Elevations above the datum are positive; elevations below it are negative. This is different from a grid coordinate (which locates a point in plan, not in height), a dimension (which measures the distance between two specific points), or a TOS/BOS notation (which gives the elevation of the top or bottom of a specific steel member). When setting anchor bolts or erecting columns, always verify the elevation against the structural drawings — not the architectural drawings, which may use a different datum.

Q38. Answer: D

Heat is the thing to be careful of. An anchor rod's strength comes from its steel chemistry and, in the higher grades, from heat treatment; taking a torch to it drives the material through temperatures that alter those properties, and flame-cutting leaves a heat-affected zone and a reduced section in a member designed to carry tension. Neither effect is visible afterward, which is what makes it dangerous - the rod looks fine and is no longer the rod the engineer designed. That is why heating and cutting are not the erector's call, and why any remedy for a misplaced rod comes back with engineering direction. The other three are the field fixes people reach for and none of them is sound. Cold bending does not restore design strength; it works the steel, leaves residual stress at the bend and puts a permanent kink where the load path wanted a straight bar. Welding an extension onto a cut rod assumes the rod steel is weldable and that a field splice can carry the design tension, and neither can be assumed. Reaming the plate hole oversize changes how load transfers between rod and plate and can leave the rod bearing on very little, which is why oversized holes at base plates carry their own plate washer requirements.

Q39. Answer: C

Ontario Regulation 213/91 s. 121 (3): "A misfired explosive load removed from an explosive actuated fastening tool shall be placed in a water-filled container on the project until the misfired explosive load is removed from the project." A misfire is not a dud - it is a live charge that failed to initiate, and it may initiate later under heat, friction or impact. Water kills that possibility and keeps the load identifiable and contained until it leaves the site. Returning it to the load container is the worst of the wrong answers, because the same Part keeps loads of different strengths separated and stored in a locked container, and a live misfire mixed back into good stock will be picked up and chambered by the next worker. Firing it into scrap puts an unpredictable charge behind a fastener with no control over where it goes. Binning it hands a live charge to whoever handles the waste. The same Part also requires the worker to be adequately trained and to carry proof of that training (s. 117 (1) and (2)), to wear adequate personal protective equipment including eye protection (s. 117 (3)), and to inspect the tool before use for cleanliness, free movement, an unobstructed barrel and defects (s. 118).

Q40. Answer: A

The binding rule is the manufacturer's figure: Alberta's OHS Code s.302(1) requires that the wire ropes, sheaves, spools and drums used in rigging have a diameter of not less than the diameter specified by the manufacturer for that use. Where no specification can be found, Alberta's published explanation guide to Part 21 gives the working minimum — unless otherwise specified by the manufacturer, the diameter of winding drums should not be less than 20 times the diameter of the rope, which reduces the likelihood of weakened rope caused by excessive bending stresses. Six or twelve times the rope diameter bends the rope far harder than it is built to take; thirty times is generous but is not the published minimum. Bending fatigue, not overload, is what usually kills a rope. Note that there is no single number for a rope running over a sheave: the minimum varies with rope construction, so read the rope manufacturer's table rather than carrying one figure in your head.

Q41. Answer: C

A lap splice is an overlapping zone where two bars run parallel: the bar ends overlap by a specified length (development length $\times$ splice factor) so that bar forces transfer through bond with the surrounding concrete.

Q42. Answer: D

4-M20 A325 means 4 bolts of M20 size (20 mm nominal diameter) meeting high-strength structural bolt requirements. Since 2016 the governing product specification is ASTM F3125 — ASTM A325 and A325M were withdrawn and consolidated into F3125 as grades, so the metric bolt shown here is properly ASTM F3125 Grade A325M, the metric grade covering sizes M12 to M36. Head markings and drawing callouts were deliberately left unchanged, which is why the drawing and the bolt head still read A325 — an apprentice who goes looking for the standalone ASTM A325 standard will find a withdrawn document. On Canadian work the connection itself is governed by CSA S16, Design and construction of steel structures, which sets the resistance of the bolts and the connection and also the rules for installing and inspecting high-strength bolted joints, including pretensioned assemblies. Work to the clause numbering in the edition of that standard referenced by your provincial building code.

Q43. Answer: D

Carbon arc gouging uses a carbon electrode and compressed air to melt and blow away metal. It is the standard method for removing defective welds, back-gouging root passes of groove welds for full penetration, and excavating weld repairs.

Q44. Answer: B

A self-retracting lifeline shortens a fall but does not remove the need for clearance below. The clearance is a sum, built the same way as for a shock-absorbing lanyard: the short free fall that happens before the brake engages, plus the deceleration distance travelled while the brake brings the worker to a stop, plus the distance from the harness D-ring down to the feet, plus a safety margin so the feet come to rest clear of the nearest surface or obstruction. Anchoring overhead rather than at foot level keeps the free-fall term small, but it never drives the total to zero. Add the terms, then compare the result against the distance actually available below the work; if the space is not there, that connector is the wrong choice for that position.

Q45. Answer: B

Porosity is gas trapped in the weld metal as it solidifies. Moisture in a low-hydrogen electrode coating and oil, paint or condensation on the base metal are the leading sources — they break down in the arc and release hydrogen and carbon monoxide into the puddle. The other faults listed produce different discontinuities: slag left between passes gives slag inclusions, too wide a root opening gives burn-through and a sagging root, and too little current gives an unstable arc with lack of fusion.

Q46. Answer: A

Camber is a bow along the length of the beam, so it can be read off the piece itself: lay it on level blocking, sight down the top flange from one end, and the curve shows which way the piece rises at midspan. Twenty-five millimetres over a long span is easy to see once you sight it and nearly invisible if you only look at the beam side on. Check the reading against the shop detail for that piece mark before it leaves the ground, then re-mark it. The wrong answers each fail for a different reason. Depth at the ends says nothing about camber, because camber is put into a beam of constant depth by cold cambering or by heat, so both ends measure the same. The mill mark records the producer, the heat number and the grade; it carries no orientation meaning. And 25 mm is not a tolerance to be shrugged off: it is a deliberate quantity the designer put in to be cancelled by deflection under load, so a beam flown upside down starts 25 mm low and finishes about 50 mm out at midspan, which shows up in the floor and in the deck bearing.

Q47. Answer: D

IHSA's structural steel chapter states the rule without qualification: no construction loads on steel joists until all bridging is installed and anchored and all joist-bearing ends are attached. A bare open-web joist is a slender member with almost no lateral stiffness. Its design capacity is a figure that assumes the bridging is there to hold the top chord in line; until it is, the joist can roll or buckle sideways under a fraction of that number. That is why half the design capacity is the wrong instinct - the design capacity is not a valid reference point yet, so no fraction of it is safe either. The same reasoning rules out the deck bundle answer: bundles are exactly the construction load the rule is about, and the same chapter requires them to be landed on fully bolted framing beside a main supporting beam at a column or wall, not on joists that are not yet braced.

Q48. Answer: B

WorkSafeBC's rigging Part puts it in one line: 'When a sling is applied to a sharp edge of a load, the edge or the sling must be protected to prevent damage to the sling' (Occupational Health and Safety Regulation, Part 15, s. 15.39). Ontario reaches the same hazard from the other side - O. Reg. 213/91 s. 172 (3) provides that no sling or similar device for rigging or hoisting made of web-type fabric or nylon shall be used if it may be cut, and s. 172 (1) (a) requires rigging to be suitable for its intended use. So the plate does not fly on that sling while the edge can cut it. Corner protectors, edge guards or softeners between the edge and the webbing are what removes the exposure.

None of the other three answers takes the edge away. Doubling the sling lays a second ply against the same edge in the same place, and the edge starts on it as soon as the load comes on. Halving the load does not help either, because a cut is not a strength problem: webbing is severed by pressure concentrated over a small radius, which happens at loads far below the rating, and once the fibres are through, the remaining section lets go without warning. Taking the load up slowly still brings the full weight of the plate onto the edge, and the sling then works against it every time the plate swings or settles.

Note what s. 15.39 is written about - a sling, not webbing alone. Chain and wire rope resist a sharp edge far better than polyester does, but the duty to protect the edge or the sling does not change with the material, so choosing a different sling is a decision about what to rig with, not a way of skipping protection at the corner.

Q49. Answer: D

A drift pin is only a tapered alignment tool for bolt holes. A spud wrench combines a wrench (typically open-end or box) on one end with a tapered shank for alignment on the other, allowing both alignment and bolt tightening in one tool.

Q50. Answer: B

The transverse and longitudinal ribs on deformed bar create mechanical interlock with the surrounding concrete, dramatically improving bond strength compared to smooth bars.

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